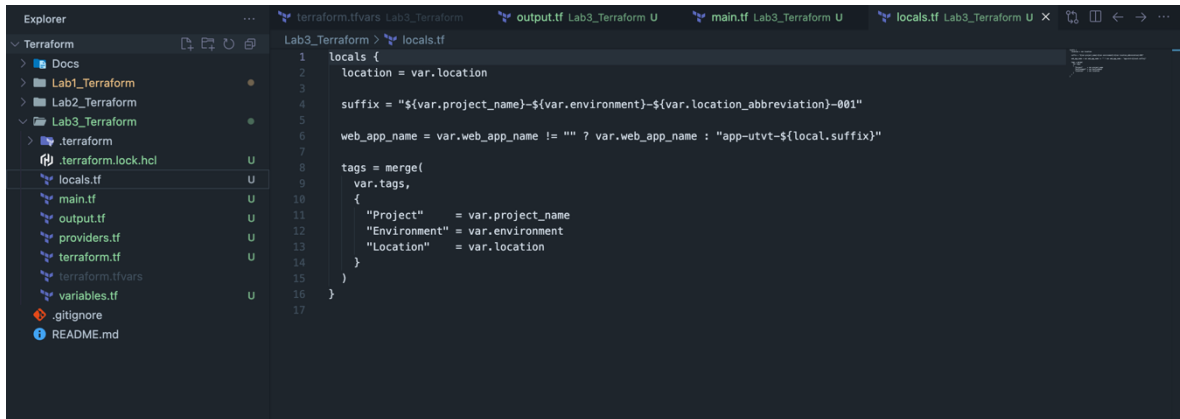


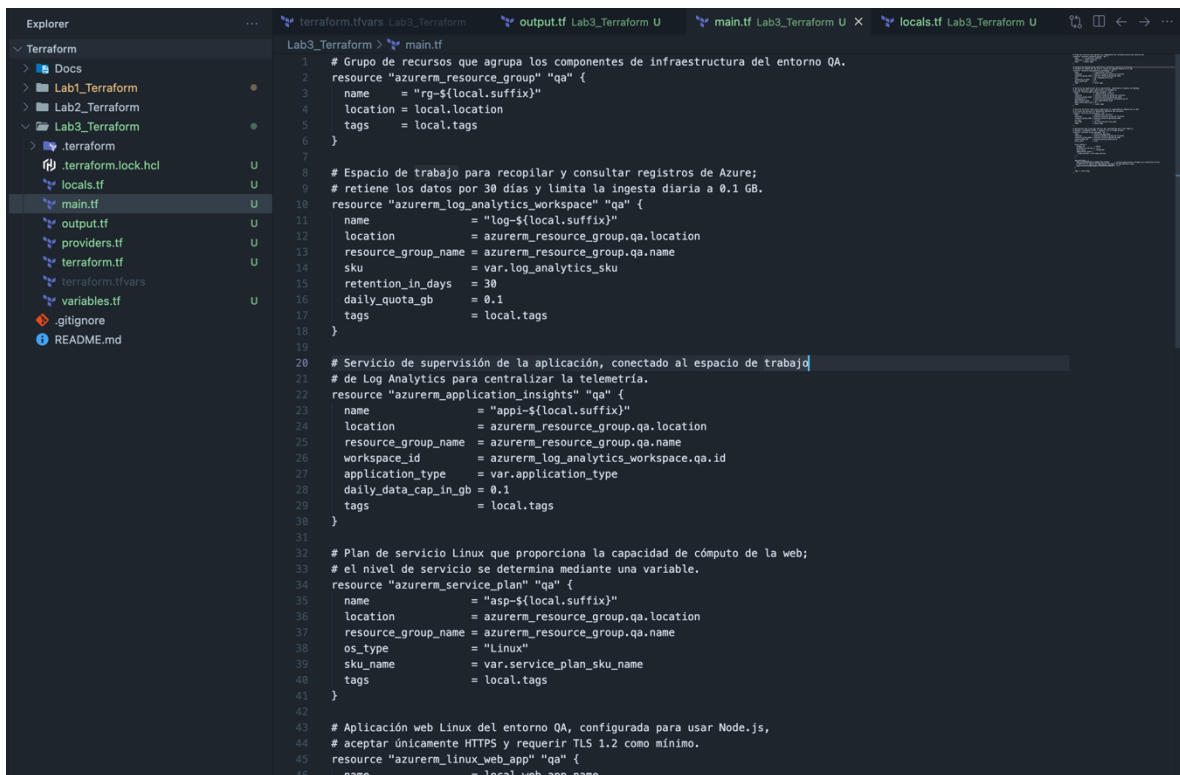
Terraform 3

Codigo



The screenshot shows the VS Code interface with the Terraform configuration for `locals.tf` in the `Lab3_Terraform` directory. The Explorer pane on the left shows the project structure, including `terraform.lock.hcl`, `locals.tf`, `main.tf`, `output.tf`, `providers.tf`, `terraform.tf`, `terraform.tfvars`, `variables.tf`, `.gitignore`, and `README.md`. The main editor displays the following HCL code:

```
1 locals {
2   location = var.location
3
4   suffix = "${var.project_name}-${var.environment}-${var.location_abbreviation}-001"
5
6   web_app_name = var.web_app_name != "" ? var.web_app_name : "app-utvt-${local.suffix}"
7
8   tags = merge(
9     var.tags,
10    {
11      "Project" = var.project_name
12      "Environment" = var.environment
13      "Location" = var.location
14    }
15  )
16 }
17
```



The screenshot shows the VS Code interface with the Terraform configuration for `main.tf` in the `Lab3_Terraform` directory. The Explorer pane on the left shows the project structure, including `terraform.lock.hcl`, `locals.tf`, `main.tf`, `output.tf`, `providers.tf`, `terraform.tf`, `terraform.tfvars`, `variables.tf`, `.gitignore`, and `README.md`. The main editor displays the following HCL code:

```
1 # Grupo de recursos que agrupa los componentes de infraestructura del entorno QA.
2 resource "azurerm_resource_group" "qa" {
3   name = "rg-${local.suffix}"
4   location = local.location
5   tags = local.tags
6 }
7
8 # Espacio de trabajo para recopilar y consultar registros de Azure;
9 # retiene los datos por 30 días y limita la ingesta diaria a 0.1 GB.
10 resource "azurerm_log_analytics_workspace" "qa" {
11   name = "log-${local.suffix}"
12   location = azurerm_resource_group.qa.location
13   resource_group_name = azurerm_resource_group.qa.name
14   sku = var.log_analytics_sku
15   retention_in_days = 30
16   daily_quota_gb = 0.1
17   tags = local.tags
18 }
19
20 # Servicio de supervisión de la aplicación, conectado al espacio de trabajo
21 # de Log Analytics para centralizar la telemetría.
22 resource "azurerm_application_insights" "qa" {
23   name = "appi-${local.suffix}"
24   location = azurerm_resource_group.qa.location
25   resource_group_name = azurerm_resource_group.qa.name
26   workspace_id = azurerm_log_analytics_workspace.qa.id
27   application_type = var.application_type
28   daily_data_cap_in_gb = 0.1
29   tags = local.tags
30 }
31
32 # Plan de servicio Linux que proporciona la capacidad de cómputo de la web;
33 # el nivel de servicio se determina mediante una variable.
34 resource "azurerm_service_plan" "qa" {
35   name = "asp-${local.suffix}"
36   location = azurerm_resource_group.qa.location
37   resource_group_name = azurerm_resource_group.qa.name
38   os_type = "Linux"
39   sku_name = var.service_plan_sku_name
40   tags = local.tags
41 }
42
43 # Aplicación web Linux del entorno QA, configurada para usar Node.js,
44 # aceptar únicamente HTTPS y requerir TLS 1.2 como mínimo.
45 resource "azurerm_linux_web_app" "qa" {
46   name = local.web_app_name
47 }
```

```
1 output "resource_group_name" {
2   description = "Nombre del Resource Group creado para QA"
3   value       = azurerm_resource_group.qa.name
4 }
5
6 output "log_analytics_workspace_id" {
7   description = "ID del espacio de trabajo de Log Analytics"
8   value       = azurerm_log_analytics_workspace.qa.id
9 }
10
11 output "application_insights_instrumentation_key" {
12   description = "Clave de instrumentacion de Application Insights"
13   value       = azurerm_application_insights.qa.instrumentation_key
14   sensitive   = true
15 }
16
17 output "service_plan_id" {
18   description = "ID del Service Plan de Azure"
19   value       = azurerm_service_plan.qa.id
20 }
21
22 output "web_app_name" {
23   description = "Nombre asignado a la aplicacion web Linux"
24   value       = azurerm_linux_web_app.qa.name
25 }
26
27 output "web_app_default_hostname" {
28   description = "URL publica generada para la Web App de QA"
29   value       = "https://${azurerm_linux_web_app.qa.default_hostname}"
30 }
31 #! for Agent
```

```
1 provider "azurerm" {
2   features {}
3   subscription_id = var.subscription_id
4 }
5
```

```
1 terraform {
2   required_providers {
3     azurerm = {
4       source  = "hashicorp/azurerm"
5       version = "~> 4.0"
6     }
7   }
8   required_version = ">= 1.5.0"
9 }
10
11
```

Explorer

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 - Lab3_Terraform
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 - terraform.tf
 - terraform.tfvars
 - variables.tf
 - .gitignore
 - README.md

```
1 subscription_id = "de8edfbf-276b-41f0-829d-ee20551b5138"
2
3 project_name = "integradora"
4
5 environment = "qa"
6
7 location = "mexicocentral"
8
9 location_abbreviation = "mxc"
10
11 service_plan_sku_name = "F1"
12
13 node_version = "20-lts"
14
15 tags = {
16   managed_by = "terraform"
17   owner      = "it"
18   cost_center = "utvt"
19 }
20
```

Explorer

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```
1 variable "subscription_id" {
2   description = "The Azure subscription ID."
3   type       = string
4   sensitive  = true
5 }
6
7 variable "project_name" {
8   description = "Nombre del proyecto"
9   type       = string
10  default    = "integradora"
11 }
12
13 variable "environment" {
14   description = "Entorno de despliegue (ej. dev, qa, prod)"
15   type       = string
16   default    = "qa"
17 }
18
19 variable "location" {
20   description = "Region de Azure para los recursos"
21   type       = string
22   default    = "mexicocentral"
23 }
24
25 variable "location_abbreviation" {
26   description = "Abreviacion de la region segun Cloud Adoption Framework"
27   type       = string
28   default    = "mxc"
29 }
30
31 variable "service_plan_sku_name" {
32   description = "SKU para el plan de servicio de App Service"
33   type       = string
34   default    = "F1"
35 }
36
37 variable "application_type" {
38   description = "Application type for the Azure Application Insights resource"
39   type       = string
40   default    = "web"
41 }
42
43 variable "node_version" {
44   description = "Node.js version for the Azure Linux Web App"
45   type       = string
46   default    = "20-lts"
47 }
```

Crearlos en la Nube de Azure.

```
carlosdc@192 Lab3_Terraform % terraform apply
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

azurerm_resource_group.qa: Creating...
azurerm_resource_group.qa: Still creating... [00m10s elapsed]
azurerm_resource_group.qa: Still creating... [00m20s elapsed]
azurerm_resource_group.qa: Creation complete after 26s [id=/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001]
azurerm_service_plan.qa: Creating...
azurerm_log_analytics_workspace.qa: Creating...
azurerm_service_plan.qa: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.qa: Still creating... [00m10s elapsed]
azurerm_service_plan.qa: Creation complete after 15s [id=/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.Web/serverFarms/asp-integradora-qa-mxc-001]
azurerm_log_analytics_workspace.qa: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.qa: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.qa: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.qa: Creation complete after 45s [id=/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.OperationalInsights/workspaces/log-integradora-qa-mxc-001]
azurerm_application_insights.qa: Creating...
azurerm_application_insights.qa: Creation complete after 4s [id=/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.Insights/components/appi-integradora-qa-mxc-001]
azurerm_linux_web_app.qa: Creating...
azurerm_linux_web_app.qa: Still creating... [00m10s elapsed]
azurerm_linux_web_app.qa: Still creating... [00m20s elapsed]
azurerm_linux_web_app.qa: Still creating... [00m30s elapsed]
azurerm_linux_web_app.qa: Still creating... [00m40s elapsed]
azurerm_linux_web_app.qa: Still creating... [00m50s elapsed]
azurerm_linux_web_app.qa: Creation complete after 51s [id=/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.Web/sites/app-utvt-integradora-qa-mxc-001]

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:

application_insights_instrumentation_key = <sensitive>
log_analytics_workspace_id = "/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.OperationalInsights/workspaces/log-integradora-qa-mxc-001"
resource_group_name = "rg-integradora-qa-mxc-001"
service_plan_id = "/subscriptions/de8edfbf-276b-41f0-829d-ee20551b5138/resourceGroups/rg-integradora-qa-mxc-001/providers/Microsoft.Web/serverFarms/asp-integradora-qa-mxc-001"
web_app_default_hostname = "https://app-utvt-integradora-qa-mxc-001.azurewebsites.net"
web_app_name = "app-utvt-integradora-qa-mxc-001"
carlosdc@192 Lab3_Terraform %
```

Microsoft Azure

Buscar recursos, servicios y documentos (G+)

Copilot

carlos.ciriaco@utvtol.edu...
UNIVERSIDAD TECNOLÓGICA DE...

InicioResource Manager

Resource Manager | Todos los recursos

Universidad Tecnológica del Valle de Toluca

Buscar

+ CrearAdministrar vistaActualizarExportar a CSVAbrir consultaAsignar etiquetasEliminarAnalizar selecciónAgregar a un grupo de serviciosNo agrupar

Resource Manager

Todos los recursos

Recursos favoritosRecursos recientesGrupos de recursosEtiquetasOrganizaciónHerramientasImplementacionesHelp

Filtrar por cualquierSuscripción es igual que todoGrupo de recursos es igual que todoTipo es igual que todoUbicación es igual que todoAgregar filtro

☐	Nombre	Tipo	Grupo de recursos	Ubicación	Suscripción
☐	app-utvt-integradora-qa-mxc-001	App Service	rg-integradora-qa-mxc-001	Mexico Central	Azure for Students
☐	appi-integradora-qa-mxc-001	Application Insights	rg-integradora-qa-mxc-001	Mexico Central	Azure for Students
☐	asp-integradora-qa-mxc-001	Plan de App Service	rg-integradora-qa-mxc-001	Mexico Central	Azure for Students
☐	log-integradora-qa-mxc-001	Área de trabajo de Log Analytics	rg-integradora-qa-mxc-001	Mexico Central	Azure for Students
☐	NetworkWatcher_mexicoentral	Network Watcher	NetworkWatcherRG	Mexico Central	Azure for Students
☐	vnet-integradora-dev-mexicoentral-001	Red virtual	rg-integradora-dev-mexicoentr...	Mexico Central	Azure for Students